IEEE ComSoc IEEE Communications Society | IEEE COMPUTER SOCIETY | IEEE Signal Processing Society | VTS | IEEE Transactions on Machine Learning in Communications and Networking

Received XX Month, XXXX; revised XX Month, XXXX; accepted XX Month, XXXX; Date of publication XX Month, XXXX; date of current version XX Month, XXXX.

Digital Object Identifier 10.1109/TMLCN.2022.1234567

Risk-Aware Learning for 5G-Advanced Handover Decisions

Gerard Grau, Ainna Yue Moreno-Locubiche, Josep Vidal, Senior Member IEEE, Margarita Cabrera-Bean, Senior Member IEEE

[1] *Dept. of Signal Theory and Communications, Universitat Politècnica de Catalunya - BarcelonaTech (UPC), 08034, Spain*

Corresponding author: Gerard Grau (email: gerard.grau.garcia@estudiantat.upc.edu).

This work covers activities of the project 6-SENSES grant PID2022-138648OB-I00, supported by MCIN/AEI/10.13039/501100011033, and by ERDF-EU.

ABSTRACT Introduced by 3GPP to optimize handover timing and execution, Lower-Layer Triggered Mobility (LTM) is a key enhancement for low-latency mobility management in 5G-Advanced networks. By executing pre-configured cell switches dynamically via fast Layer 1 / Layer 2 (L1/L2) channel measurements, LTM substantially reduces user-plane interruption time. However, making fast handover (HO) decisions under rapid multipath channel fading poses severe reliability risks, where suboptimal decisions lead to costly handover failures (HOF) and radio link failures (RLF). While standard deep reinforcement learning (RL) such as Deep Q-Networks (DQN) achieves higher spectral efficiency and greater signaling frugality than legacy 3GPP heuristics, we uncover a critical *out-of-domain generalization gap*: scalar DQN fits the local channel profiles of the training trajectories and suffers an almost two-fold increase in failure rate (from 0.67 to 1.12 HOF/min) when evaluated on unseen channel fading realizations. To overcome this fundamental limitation, we propose a risk-aware distributional reinforcement learning framework based on Quantile Regression DQN (QR-DQN). We show that learning the full return distribution across $N = 25$ quantiles acts as a powerful regularizer, restoring out-of-domain generalization. Furthermore, we reveal that while naive tail truncation (training only on the worst-case quantiles) suffers from severe tail-overfitting and gradient starvation, a smooth two-step spectral distortion (*Soft-step*) applied to the predicted return quantiles during action selection safely guides policies toward risk aversion while preserving full distributional training across all quantiles. Evaluated across 1,000 completely unseen mobile user trajectories over 5 independent seeds, Soft-step achieves the lowest failure rate in the benchmark (0.74 HOF/min), halves control-plane signaling overhead compared to 3GPP heuristics (5.8 vs. 11.3 HO/min), and preserves full spectral efficiency (4.04 bps/Hz), avoiding the 7% throughput penalty incurred by contextual bandits.

INDEX TERMS 5G-Advanced, lower-layer triggered mobility, distributional reinforcement learning, quantile regression, out-of-domain generalization, risk-sensitive control, conditional value at risk.

I. Introduction

Fifth-generation (5G) and 5G-Advanced radio access networks deploy dense, multi-sector cellular topologies operating at higher frequency bands. As user equipments (UEs) move through these dense networks, they cross cell and sector boundaries frequently, necessitating rapid and reliable mobility management. Each handover (HO) is a high-stakes reconfiguration: executed properly, it sustains peak spectral efficiency and guarantees continuous connectivity; executed poorly or tardily, it triggers handover failures (HOF), radio link failures (RLF), or wasteful "ping-pong" (PP) oscillations back to the previously serving cell.

To address the latency limitations of legacy Radio Resource Control (RRC) signaling, 3GPP Release 18 introduced *Lower-Layer Triggered Mobility* (LTM) [1]. LTM decouples candidate cell preparation from execution by pre-configuring candidate target cells at higher layers and executing the cell switch dynamically via fast Medium Access Control (MAC) control elements (CE) driven by Layer 1 / Layer 2 (L1/L2) channel measurements. While LTM min-

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/

imizes interruption time, it shifts the operational challenge to making cell selection decisions under fast-fading channel variations without the smoothing benefit of slow Layer 3 filtering. That selection is the decision problem we study.

Prior learning-based approaches have cast cell selection as heuristic rules, contextual multi-armed bandits (CMAB), or surveyed various reinforcement learning formulations for handover optimization [2]. While myopic bandit formulations improve safety compared to rigid 3GPP thresholds, their one-step objective optimizes only immediate rewards (such as momentary channel gains), rendering them unable to anticipate future signal degradation and causing a notable throughput capacity penalty. Conversely, standard Reinforcement Learning (RL) techniques like Deep Q-Networks (DQN) [3] optimize long-term cumulative discounted returns. However, scalar DQN learns only the expected return, which fails to account for the dispersion and extreme tail risks inherent to wireless fading channels.

Crucially, in this paper we uncover a major vulnerability in applying standard deep RL to wireless mobility: the **Out-of-Domain Generalization Gap**. When scalar DQN is trained and tested on the same channel realizations, it appears to perform well. Yet, when evaluated on completely unseen mobility trajectories, its handover failure rate nearly doubles. Scalar value learners overfit to local fading variations rather than discovering generalizable transition dynamics.

To solve this challenge, we introduce a risk-aware distributional reinforcement learning framework:

- We formulate 5G LTM handover as a Markov Decision Process (MDP) and implement a rigorous out-of-domain evaluation protocol, training on 2,000 UE trajectories and validating greedily on 1,000 disjoint, unseen test trajectories.
- We demonstrate that modeling the full return distribution using Quantile Regression DQN (QR-DQN) [4] provides a rich training signal that acts as an implicit regularizer, fully closing the scalar generalization gap.
- We investigate risk-aware policies and expose the pitfall of *Truncated Risk* (learning only the tail), which leads to policy instability and tail-overfitting. Similarly, strict Conditional Value at Risk (Hard CVaR) induces a "policy paralysis" trap where the agent over-reserves resources but hesitates to switch cells.
- We propose **Soft-step QR-DQN**, a spectral risk formulation that trains on the complete return distribution while smoothly upweighting the lower tail quantiles during action selection. By adjusting the tail fraction and distortion ratio, Soft-step enables flexible risk-sensitive control that achieves the lowest failure rate in the benchmark, halves control-plane signaling overhead compared to 3GPP heuristics, and preserves peak spectral efficiency without the throughput penalty of contextual bandits.

The remainder of this paper is organized as follows. Section II provides the essential background on 5G-Advanced LTM and reviews related work in distributional deep reinforcement learning for wireless applications. Section III details the 5G-Advanced cellular system model and formalizes the LTM handover decision problem as a Markov Decision Process (MDP). Section IV introduces the proposed QR-DQN framework alongside the Soft-step risk distortion strategy for risk-averse policy learning. Section V outlines the experimental setup, simulation parameters, and out-of-domain evaluation protocols. Section VI presents the quantitative performance evaluation, throughput gap analysis, and generalization results across unseen multipath fading trajectories. Finally, Section VII concludes the paper and outlines future research directions.

II. Background and Related Work

A. LTM and Handover Optimization

In legacy mobility the network reconfigures a UE's serving cell through higher-layer (RRC) signaling driven by filtered network-protocol L3 measurements—robust but slow [1]. *Lower-layer triggered mobility* (LTM) instead prepares one or more candidate cells in advance and lets a fast L1/L2 measurement event trigger the execution of a pre-configured switch, reducing the interruption time but tightening the decision loop. The operational cost of a mobility policy is captured by a small set of events: a *handover* (HO) whenever the serving cell changes, a *ping-pong* (PP) when the UE returns to a recently left cell within a short window, a *handover failure* (HOF) when the target link cannot be established, and a *radio-link failure* (RLF) when the serving link collapses after a sustained run of out-of-sync observations within a time window. A good policy minimizes HOF and PP while sustaining spectral efficiency.

A prior learning-based work optimizes this trade-off with heuristics over an expected-KPI objective. The work in [5] optimizes network decisions by maximizing an expected objective using contextual bandits. To further enhance signal quality and target pre-selection, this framework is optionally augmented with Reconfigurable Intelligent Surfaces (RIS) and LMMSE channel prediction, showcasing how physical-layer enhancements can be stacked on top of decision-making layers to stabilize vehicular or cellular links. While this represents the closest baseline to our work, it remains strictly risk-neutral. Our proposed framework directly addresses this limitation by introducing risk-awareness into the decision-making process.

B. Distributional Reinforcement Learning

Distributional RL predicts the full distribution of the return rather than its expectation. Instead of modeling the scalar action-value $Q(x, a)$ for each state x and action a, the agent models the return as a random variable $Z(x, a)$ that obeys the distributional Bellman equation:

$$Z(x, a) \stackrel{D}{=} R(x, a) + \gamma\, Z(X', A'), \tag{1}$$

where $\stackrel{D}{=}$ denotes equality in distribution, $R(x,a)$ is the reward, γ the discount factor, and (X',A') the (random) next state–action pair. Even when the policy ultimately acts on the mean $\mathbb{E}[Z(x,a)]$, learning the distribution provides a richer per-update signal and consistently improves the resulting policy. Most practical distributional RL algorithms approximate the cumulative distribution as a staircase function; representative methods include C51 [6], QR-DQN [4], IQN [7] and FQF [8]. We build on QR-DQN, which parametrizes the distribution by estimating the locations of a fixed set of quantiles. Unlike Implicit Quantile Networks (IQN) [7] or Fully Parameterized Quantile Function (FQF) [8], which introduce auxiliary quantile proposal networks and require run-time stochastic sampling, QR-DQN operates on a fixed, deterministic set of quantiles. This yields a closed-form quantile Huber loss and deterministic inference with minimal computational latency and footprint, a critical requirement for high-cadence 5G L1/L2 LTM decision loops.

C. Risk-Sensitive Reinforcement Learning

In standard distributional RL the greedy action maximizes the expected return $\mathbb{E}[Z(x,a)]$. In vanilla QR-DQN with N midpoint quantiles $\theta_i(x,a)$ per each state-action pair (x,a), this expectation is approximated by the uniform average of the N predicted quantiles $\frac{1}{N}\sum_{i=1}^{N}\theta_i(x,a)$, which represents the risk-neutral special case. Intuitively, each quantile $\theta_i(x,a)$ specifies the exact position on the horizontal axis of the return distribution corresponding to a fixed probability checkpoint when action a is executed at state x. One may instead aggregate the θ_i under a different weighting criterion to obtain a risk-sensitive policy. Selecting an action from a set of competing return distributions is fundamentally nontrivial because distributions lack a canonical total order—under stochastic dominance, many pairs are incomparable. Any scalar aggregation must therefore impose an explicit ordering, directly encoding a specific risk attitude.

A common risk-averse choice replaces the expectation with the Conditional Value at Risk at level α, CVaR_α, defined as the average of the worst α-fraction of outcomes. Here $\alpha \in (0,1]$ is the tail fraction that is integrated, not an individual quantile position. More generally, replacing the uniform average by non-negative quantile weights $\sum_{i=1}^{N}\omega_i\theta_i(x,a)$ with $\sum_{i=1}^{N}\omega_i = 1$ yields a *spectral* (or distortion) risk measure [9], with the risk-neutral expectation ($\omega_i = 1/N$) and CVaR_α ($\omega_i = \mathbb{I}\{i \le k\}/k$, with $k = \lceil \alpha N \rceil$ such that $\alpha \approx k/N$) as specific piecewise-uniform instances. This weighted-quantile framework defines the design space for the risk-aware policies we study.

CVaR and related risk-sensitive criteria have a rich history in optimization and control: Rockafellar and Uryasev [10] established convex optimization techniques for CVaR; Chow and Ghavamzadeh [11] formulated CVaR objectives for Markov Decision Processes; and Tamar *et al.* [12] developed policy gradient methods for risk-averse evaluation. Distributional RL makes evaluating such criteria particularly straightforward and computationally direct, since the return quantiles needed to compute CVaR_α and spectral measures are predicted explicitly by a neural network without requiring secondary sampling loops.

D. Distributional RL in Wireless Communications

While the initial successes of distributional RL focused primarily on arcade games, autonomous driving, and robotics, recent literature has successfully deployed distributional frameworks to address optimization challenges within next-generation wireless networks. In [13], a trajectory planning problem in drone networks is addressed, by using the Conservative Q-Learning (CQL) algorithm. A regularization term is introduced to the DQN objective function to reduce overestimation bias in offline learning by penalizing discrepancies between estimated returns across all actions and the values recorded in the dataset. In [14], CQL is combined with Quantile Regression (QR-DQN) for unmanned aerial vehicle (UAV) trajectory optimization. In [15], quantile regression deep Q-networks are deployed to learn the complete cumulative distribution of end-to-end delays rather than simple averages, optimizing routing decisions under deadline constraints and preventing traffic bottlenecks.

To the best of our knowledge, our work is the first to investigate risk-aware distributional RL for 5G-Advanced lower-layer mobility decisions. Furthermore, in contrast to prior art, this paper focuses on out-of-domain generalization across fading profiles and the downstream distortion of predicted quantiles to prevent tail failures.

III. System Model and Problem Formulation

A. 5G LTM Deployment and Simulator

We consider a cellular deployment of $N_{\mathrm{BS}} = 7$ tri-sector base stations (21 sectors) serving mobile UEs moving along urban trajectories at vehicular speeds. Channel gains between sectors and UEs incorporate distance-dependent path loss, shadow fading, and small-scale multipath fast fading generated from calibrated ray-tracing and geometric models. The simulator advances at a 10 ms tick resolution: at each step it derives the serving signal-to-interference-plus-noise ratio (SINR) from channel gains, transmit power ($P_{\mathrm{tx}} = 25$ dBm), and inter-cell interference, mapping it through a modulation-and-coding scheme (MCS) to spectral efficiency. An SINR below the outage threshold (-3 dB) yields zero bit rate, and a sustained run of out-of-sync indications escalates to a radio-link failure (RLF). System parameters (Table 1) follow the calibrated LTM reference implementation of [5].

LTM maintains a small set of *prepared* candidate cells (up to $N_{\mathrm{prep}}^{\max} = 5$), selected from the strongest neighbors whose RSRP falls within a preparation margin of $\Delta_{\mathrm{prep}} = -3$ dB relative to the serving cell ($\mathrm{RSRP}_{\mathrm{cand}} \ge \mathrm{RSRP}_{\mathrm{serv}} + \Delta_{\mathrm{prep}}$). In the standard heuristic, a prepared cell is executed when its RSRP exceeds the serving cell by an execution margin ($H_{\mathrm{exec}} = 3$ dB). We replace this rigid execution trigger with

our learned agent: at each decision step, the agent chooses whether to stay or execute an LTM switch to a prepared cell.

TABLE 1. Calibrated System and Simulation Parameters

Parameter	Value
Base stations × sectors	$7 \times 3 = 21$
Training trajectories	2,000 (300 s each)
Evaluation trajectories (unseen)	1,000 (300 s each)
Simulator step / RL cadence	10 ms / 100 ms event-driven
Transmit power (P_{tx})	25 dBm
Thermal noise floor (σ^2)	-101 dBm (20 MHz bandwidth)
Receiver sensitivity threshold	-95 dBm
Outage (RLF) SINR threshold	-3 dB
RLF threshold ($N_{\text{OOS}}^{\text{RLF}}$)	20 consecutive ticks (200 ms)
Preparation offset (Δ_{prep})	-3 dB
Max. prepared candidate cells	5
Execution offset (LTM trigger)†	3.0 dB
Execution time (T_{exec})	40 ms

†The +3 dB execution offset is the LTM baseline's hard handover trigger; the RL agent replaces it with a learned, value-based decision.

B. MDP Formulation

We model LTM HO as a Markov decision process in which the agent acts only at handover-decision opportunities. These arise once per LTM cycle and are unevenly spaced in time; we index the agent's successive decisions by t and treat each as one step.

State. An 88-dimensional vector: UE speed (1) and serving-cell tenure time (1), the one-hot serving sector (21), per-sector RSRP (21), moving-average MCS (21) and moving-average SINR (21), and the UE position (x, y) (2).

Action. At each decision the agent chooses a target from the action set $\{\text{serving}\} \cup \{\text{prepared}\}$, at most six entries, since preparation caps the candidate list at 5. Choosing the serving cell is a *stay* (no handover); choosing a non-serving prepared cell executes a handover to it. Because the agent itself fires the handover, it replaces LTM's +3 dB execution trigger with the learned value of acting now versus staying, while preparation and the candidate restriction remain the reference heuristic methodology of [1].

Reward. Each decision is rewarded over the interval until the next decision by the multiplicative reward of [5],

$$R_t = R_t^{\text{thr}} \cdot \frac{\alpha_{\text{HO}}^{\mathbb{I}_t^{\text{HO}}} \, \alpha_{\text{PP}}^{\mathbb{I}_t^{\text{PP}}} \, \alpha_{\text{HOF}}^{\mathbb{I}_t^{\text{HOF}}}}{1 + e^{\,2\,(N_t^{\text{OOS}} - 2)}}, \tag{2}$$

where R_t^{thr} is the throughput term—the serving spectral efficiency (bps/Hz) *averaged over the decision interval*; the indicators $\mathbb{I}_t^{\text{HO}}, \mathbb{I}_t^{\text{PP}}, \mathbb{I}_t^{\text{HOF}} \in \{0, 1\}$ flag whether a handover, ping-pong or handover failure occurred *anywhere within* that interval; and $\alpha_{\text{HO}}, \alpha_{\text{PP}}, \alpha_{\text{HOF}} \in (0, 1]$ are the corresponding multiplicative penalties (defaults α_{HOF}=0.1, α_{HO}=0.8, α_{PP}=0.9, so a failure is penalized far more harshly than a routine handover). The final factor is a reverse sigmoid of the out-of-sync counter N_t^{OOS} (the number of consecutive out-of-sync indications on the serving link): it stays near 1 while the link is in sync, equals 0.5 at N_t^{OOS}=2, and decays toward 0 as the link approaches radio-link failure. As the penalty factors lie in $(0, 1]$, the reward formulation discounts the ideal throughput R_t^{thr} according to the penalties triggered by mobility failures and link degradation, which the distributional agent models as a random return.

Because decisions are event-driven, a policy's own choices change how many actions it faces per episode; we therefore compare policies by their mean reward *per decision* rather than the episode sum, which would otherwise credit a policy merely for being consulted more often (see details in Section V).

IV. Method

A. QR-DQN and the Quantile Loss

QR-DQN parametrizes $Z(x, a)$ by its inverse Cumulative Distribution Function (CDF) at N quantile fractions $\tau_1, \ldots, \tau_N \in [0, 1]$, learning $\theta_i(x, a) \approx F^{-1}_{Z(x,a)}(\tau_i)$. These $\theta_i(x, a)$ are the outputs of a neural network: a shared trunk maps the state x to an output layer (the *head*) that emits the N quantile estimates for every action a, and the network parameters ϕ are fit by minimizing a loss function as explained hereafter and defined in (6). Vanilla QR-DQN partitions the cumulative probability interval $[0, 1]$ into N equal bins of width $\Delta\tau_i = 1/N$, placing the quantile fractions at the bin midpoints $\tau_i = (i - \frac{1}{2})/N$. The aggregated value used for action selection,

$$\mathbb{E}[Z(x, a)] = \int_0^1 F^{-1}_{Z(x,a)}(\tau)\, d\tau \;\approx\; \sum_{i=1}^{N} w_i\, \theta_i(x, a), \tag{3}$$

is a finite-sample Riemann quadrature. Because the bin widths $\Delta\tau_i = 1/N$ are constant across the entire support, the corresponding quadrature weights are exactly uniform: $w_i = \Delta\tau_i = 1/N$ ($\sum_i w_i = 1$). The pair (τ_i, w_i) defines the *quantile scheme*, which naturally recovers the sample average for risk-neutral expectation. Risk sensitivity is introduced in Section B by altering this aggregation or restricting the target quantiles; the loss below keeps the same form throughout.

The quantiles are trained with the quantile (pinball) Huber loss. Writing the Huber function with threshold κ as

$$\mathcal{H}_\kappa(u) = \begin{cases} \frac{1}{2}u^2, & |u| \le \kappa, \\ \kappa\big(|u| - \frac{1}{2}\kappa\big), & |u| > \kappa, \end{cases} \tag{4}$$

the quantile-Huber penalty at fraction τ is given by

$$\rho_\tau^\kappa(u) = \big|\tau - \mathbb{I}\{u < 0\}\big|\, \mathcal{H}_\kappa(u). \tag{5}$$

The Huber threshold κ rounds the kink of the plain pinball (absolute-value) loss into a quadratic region near zero, giving bounded continuous gradients that are robust to the large bootstrapped errors common early in training [4]. The asymmetric prefactor $|\tau - \mathbb{I}\{u < 0\}|$ is what makes this *quantile* regression rather than ordinary regression: it penalizes errors asymmetrically—weight τ on one side of zero and $1 - \tau$ on

the other—so the per-quantile loss is minimized when θ_i sits at the τ_i-quantile of its target rather than at the mean. The per-transition loss sums the pairwise temporal-difference errors between each of the N *predicted* quantiles (the online network's current outputs) and each of the N *target* quantiles $y_j = r + \gamma\,\theta_j(x', a^\star)$ (bootstrapped from a frozen target network), where $a^\star$ is the greedy action under the aggregated value (3). The resulting quantile loss function, given in (6),

$$\mathcal{L}(x,a) = \frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{N}\rho_{\tau_i}^{\kappa}\big(y_j - \theta_i(x,a)\big), \tag{6}$$

is best understood through the theoretical foundation established by Dabney *et al.* [4]. Ideally, each prediction $\theta_i(x,a)$ would minimize the expected quantile-Huber loss against the random target $Y = r + \gamma Z(x', a^\star)$, whose true minimizer is the τ_i-quantile of Y. Because Y is represented empirically by its N bootstrapped target particles $\{y_j\}_{j=1}^{N}$, projecting the distributional Bellman operator under the 2-Wasserstein metric requires regressing each predicted quantile θ_i against *all* N target quantiles $\{y_j\}$. A naive one-to-one regression between identical indices ($i = j$) would falsely presume that quantile ordering is preserved under stochastic transitions and additive rewards. Regressing every predicted quantile against all N targets ensures that the particles correctly capture the true distribution of $r + \gamma Z(x', a^\star)$ without requiring sorted matching.

B. Risk-Aware Quantile Aggregation

The quantile loss function (6) fixes how the return distribution is learned but leaves the aggregation (3) free. A risk-sensitive policy keeps the same network and replaces the uniform average by a non-negative weighting

$$Q_\omega(x,a) = \sum_{i=1}^{N}\omega_i\,\theta_i(x,a), \quad \omega_i \geq 0, \quad \textstyle\sum_{i=1}^{N}\omega_i = 1, \tag{7}$$

acting greedily on Q_ω. Note that the aggregation weights ω_i directly influence the selection of the optimal action $a^\star$, upon which the target labels y_j in (6) depend. To achieve a risk-averse measure, the weights ω_i must be monotonically non-increasing in τ_i. This yields a *spectral* (distortion) risk measure as explained in Section C, with the expectation as its uniform special case (Risk-Neutral QR-DQN).

We inject risk aversion into the policy via two distinct mechanisms: first, by *re-weighting* the estimated quantiles of a faithfully trained return distribution—exemplified by Hard CVaR and Soft-step policies, with the standard mean serving as the risk-neutral baseline—which leaves (6) untouched and introduces zero training overhead; and second, by *truncating or isolating* the quantiles to focus exclusively on the tail during training (Truncated CVaR and single-quantile policies).

To provide a rigorous, systematic comparison, we formalize the evaluated decision policies and their exact neural output dimensions:

1) **3GPP Baseline (LTM):** Standard 3GPP Release 18 A3 event condition triggering an LTM handover when a prepared candidate sector's RSRP exceeds the serving sector's RSRP by hysteresis $H_{\text{exec}} = 3.0\,\text{dB}$ (unlearned heuristic, zero neural parameters).
2) **Contextual Bandit (CMAB):** Myopic ($\gamma = 0$) contextual multi-armed bandit following Moreno-Locubiche *et al.* [5], optimizing immediate expected reward from L1/L2 measurements.
3) **Scalar DQN:** Conventional Deep Q-Network estimating expected return $\mathbb{E}[R]$, featuring an output layer of $|A| = 21$ units with greedy action selection $\arg\max_a Q(s,a)$.
4) **Risk-Neutral QR-DQN (RN QR-DQN):** Full $N = 25$ quantile regression network featuring an output layer of $|A| \times N = 21 \times 25 = 525$ units. Actions are chosen via the expected return with uniform weights $\omega_i = 1/N$.
5) **Hard CVaR QR-DQN ($\alpha \in \{0.10, 0.25\}$):** Employs the full $N = 25$ network (525 output units), as in Risk-Neutral QR-DQN, but selects actions by averaging exclusively the lower $k = \lceil \alpha N \rceil$ quantiles:
$$\omega_i^{\text{CVaR}} = \frac{\mathbb{I}\{i \leq k\}}{k}, \quad k = \lceil \alpha N \rceil. \tag{8}$$
6) **Truncated QR-DQN ($k \in \{1, 3, 7\}$):** A tail-restricted architecture that trains *only* $k = \lceil \alpha N \rceil$ quantiles located in $[0, \alpha]$, reducing the network output layer to $|A| \times k$ units ($21 \times 7 = 147$ units for $k = 7$; 63 units for $k = 3$; and 21 units for $k = 1$). It minimizes the restricted quantile Huber loss:
$$\mathcal{L}_k(x,a) = \frac{1}{k}\sum_{i=1}^{k}\sum_{j=1}^{k}\rho_{\tau_i}^{\kappa}\big(y_j - \theta_i(x,a)\big). \tag{9}$$
When $k = 1$ ($\tau = \alpha/2 = 0.125$), this instantiates the dedicated **Risk-Averse 1-Quantile (RA-1q)** agent with $|A| = 21$ outputs. Unlike post-hoc evaluation of a single quantile from a full model, RA-1q is trained from scratch with gradients backpropagating solely through a single quantile $\tau = 0.125$.
7) **Soft-step QR-DQN (Proposed, $\alpha = 0.25, r = 2.0$):** A smooth spectral risk policy using the full $N = 25$ network (525 output units) to retain complete distribution learning across training, while reweighting the lower $k = \lceil \alpha N \rceil$ quantiles by distortion factor $r \geq 1$ at decision time:
$$\omega_i(r) = \frac{r\,\mathbb{I}\{i \leq k\} + \mathbb{I}\{i > k\}}{r\,k + (N - k)}. \tag{10}$$
Here, the distortion parameter $r \geq 1$ controls the relative emphasis assigned to adverse tail outcomes: setting $r = 1$ recovers the standard risk-neutral expectation, while $r > 1$ amplifies the decision weight of the lower tail quantiles relative to the distribution body. For instance, with $\alpha = 0.25$ ($k = 7$) and $r = 2.0$, tail quantiles below the first quartile receive twice the relative weight of the remaining quantiles during action selection.

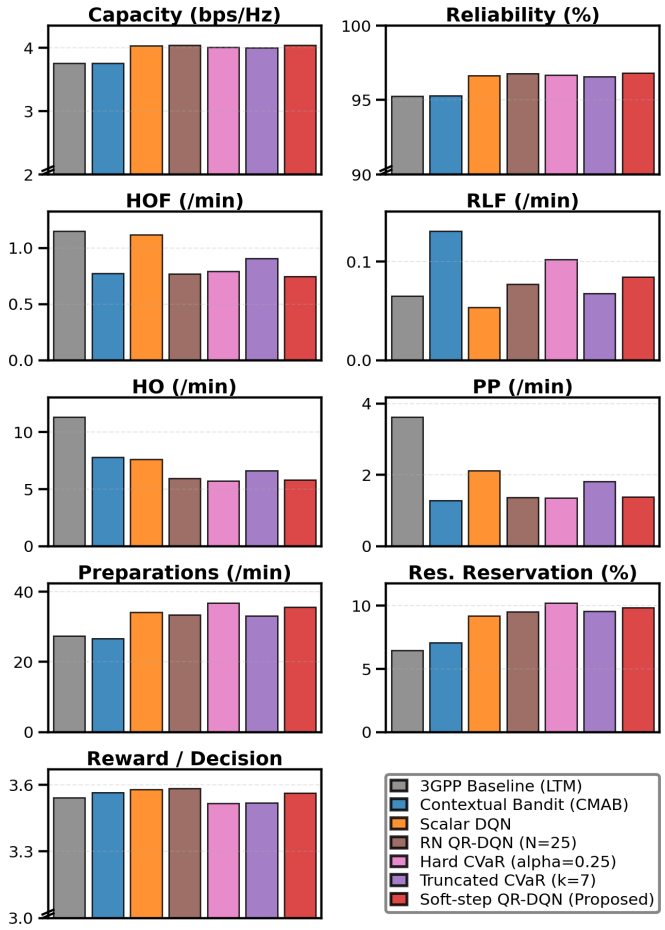


FIGURE 1. Comprehensive 9-KPI out-of-domain generalization benchmark on 1,000 unseen test trajectories over 5 seeds. The 5×2 single-column layout evaluates all policies across the 9 standardized operational KPIs, with the 10th panel providing the unified algorithm legend.

V. Experimental Setup and Evaluation Protocol

A. Rigorous Out-of-Domain Evaluation Protocol

To assess true policy generalization under realistic propagation dynamics, we implement a strict out-of-domain evaluation methodology. Because wireless channels exhibit rapid, location-specific fast fading, evaluating policies on unseen mobility trajectories is essential to verify that the agent learns generalizable transition dynamics rather than memorizing local channel profiles.

Our evaluation environment separates mobile trajectories into disjoint datasets:

- **Training Split:** 2,000 independent UE trajectories, each 300 s long at 10 ms physical simulation resolution, totaling 600,000 s of mobile operation (60×10^6 physical simulation steps).
- **Evaluation Split (Unseen):** 1,000 completely separate trajectories, totaling 300,000 s of mobile operation (30×10^6 physical simulation steps).

Each learned model is trained for 6,000 episodes, ensuring three complete passes over the training trajectories. Post-training evaluation is conducted strictly greedily ($\epsilon = 0.0$) with frozen network weights across all 1,000 test trajectories. To ensure statistical significance, all experiments are replicated across 5 independent random seeds (pseudo-random generator seeds 42–46 in PyTorch and NumPy, which govern neural network weight initialization, replay buffer batch sampling, and environment exploration actions during training), and we report cross-seed means alongside standard deviations.

B. Physical Key Performance Indicators (KPIs)

In accordance with 3GPP standards and the benchmark formulation of [5], we evaluate 8 physical operational KPIs alongside the primary optimization reward:

1) **Handover Failure (HOF) Rate (/min):** Frequency of failed handover executions where target channel SINR drops below the outage threshold (-3 dB) during the 40 ms execution window.
2) **Average Spectral Efficiency / Capacity (bps/Hz):** Downlink spectral efficiency mapped through 3GPP Modulation and Coding Schemes (MCS).
3) **Handover (HO) Rate (/min):** Frequency of cell switches executed per minute of user connection.
4) **Ping-Pong (PP) Rate (/min):** Unnecessary rapid cell reversals where a UE returns to its previous serving sector within a short critical window (1 s).
5) **Radio Link Failure (RLF) Rate (/min):** Rate of complete link drops caused by sustained out-of-sync conditions ($N^{\text{OOS}} \geq 20$).
6) **Cell Preparation Rate (/min):** Rate of candidate target cell preparations initiated by L3 RSRP filtering.
7) **Resource Reservation (%):** Percentage of physical downlink resources pre-reserved by base stations to maintain prepared candidate contexts.
8) **Link Reliability (%):** Percentage of active transmission ticks where the UE maintains non-outage connectivity (SINR ≥ -3 dB).

C. Model Architecture and Shared Hyperparameters

All deep RL agents employ an identical Multi-Layer Perceptron (MLP) trunk consisting of two hidden layers with 512 units each and ReLU activations, ensuring that performance differences stem entirely from the learning objectives and decision policies rather than representation capacity. Network hyperparameters are summarized in Table 3. During training, the agent performs one mini-batch gradient descent step (using a mini-batch size of 256 transitions sampled uniformly from the replay buffer of 10^5 steps) every 8 environment steps (decision intervals). Exploration follows an exponential ϵ-greedy schedule decaying from 1.0 to 0.05. Target networks are updated softly at each environment step

TABLE 2. Comprehensive 9-KPI Out-of-Domain Generalization Benchmark (5 Seeds, 1,000 Unseen Test Trajectories, $\epsilon = 0.0$). Bold indicates the best result; underline indicates the second-best.

Method	Throughput ↑ (bps/Hz)	Reliability ↑ (%)	HOF Rate ↓ (/min)	RLF Rate ↓ (/min)	HO Rate ↓ (/min)	Ping-Pong ↓ (/min)	Prep. Rate ↓ (/min)	Reservation ↓ (%)	Reward / Dec. ↑
3GPP Baseline (LTM)	3.75 ± 0.29	$95.2\% \pm 1.3\%$	1.15 ± 0.69	$\underline{0.07 \pm 0.11}$	11.3 ± 3.0	3.6 ± 1.5	$\underline{27.3 \pm 7.1}$	$\mathbf{6.4\% \pm 0.4\%}$	3.52 ± 0.30
Contextual Bandit (CMAB) [5]	3.76 ± 0.33	$95.3\% \pm 1.8\%$	0.77 ± 0.45	0.13 ± 0.27	7.8 ± 2.1	$\mathbf{1.3 \pm 0.8}$	$\mathbf{26.7 \pm 8.3}$	$\underline{7.1\% \pm 0.9\%}$	3.57 ± 0.33
Scalar DQN	4.03 ± 0.01	$96.7\% \pm 0.1\%$	1.12 ± 0.17	$\mathbf{0.05 \pm 0.01}$	7.6 ± 0.8	2.1 ± 0.5	34.2 ± 2.1	$9.2\% \pm 0.2\%$	$\underline{3.58 \pm 0.07}$
RN QR-DQN ($N = 25$)	$\mathbf{4.04 \pm 0.00}$	$\underline{96.8\% \pm 0.1\%}$	$\underline{0.77 \pm 0.15}$	0.08 ± 0.00	5.9 ± 0.7	1.4 ± 0.4	33.4 ± 1.5	$9.5\% \pm 0.0\%$	$\mathbf{3.58 \pm 0.03}$
Hard CVaR QR-DQN ($\alpha = 0.25$)	4.01 ± 0.01	$96.7\% \pm 0.1\%$	0.79 ± 0.10	0.10 ± 0.02	$\mathbf{5.7 \pm 0.8}$	$\underline{1.4 \pm 0.5}$	36.9 ± 2.0	$10.2\% \pm 0.1\%$	3.52 ± 0.04
Truncated QR-DQN ($k = 7$)	4.00 ± 0.01	$96.6\% \pm 0.1\%$	0.90 ± 0.21	0.07 ± 0.01	6.6 ± 1.1	1.8 ± 0.6	33.1 ± 2.7	$9.6\% \pm 0.3\%$	3.52 ± 0.09
Soft-step QR-DQN (Proposed)	$\underline{4.04 \pm 0.01}$	$\mathbf{96.8\% \pm 0.1\%}$	$\mathbf{0.74 \pm 0.07}$	0.08 ± 0.01	$\underline{5.8 \pm 0.5}$	1.4 ± 0.3	35.7 ± 1.3	$9.8\% \pm 0.2\%$	3.56 ± 0.06

with an update rate factor of 0.01. The Huber loss threshold is fixed at $\kappa = 0.25$.

TABLE 3. Shared Neural Network and Training Hyperparameters

Hyperparameter	Value
Trunk architecture	2-layer MLP (512×512, ReLU)
Optimizer	Adam
Learning rate	10^{-4}
Discount factor (γ)	0.9
Replay buffer capacity	10^5 transitions
Mini-batch size	256
Training frequency	8 environment steps
Target network update	Soft update, $\tau = 0.01$
Exploration schedule	$\epsilon : 1.0 \rightarrow 0.05$ ($\times 0.995$/episode)
Quantile count (N)	25 (Midpoint scheme)
Huber threshold (κ)	0.25
Training episodes	6,000 (3 complete passes per trajectory)
Independent seeds	5 (Seeds $42, 43, 44, 45, 46$)
Evaluation mode	Greedy ($\epsilon = 0.0$), frozen weights

VI. Performance Evaluation and Generalization Analysis

Before examining individual policy behaviors, we outline the structure of the evaluation benchmark. Table 2 and Fig. 1 present the complete out-of-domain evaluation across all 9 physical operational KPIs over the 1,000 unseen test trajectories, averaged across 5 independent training seeds (42–46). The benchmark is structured into three tiers: legacy and myopic baselines (3GPP heuristic and CMAB), value-based scalar deep RL (DQN), and the distributional QR-DQN family spanning risk-neutral expectation, tail truncation ($k \in \{1, 3, 7\}$), Hard CVaR ($\alpha \in \{0.10, 0.25\}$), and our proposed Soft-step formulation.

A. The Out-of-Domain Generalization Gap in Deep RL

When evaluated on the training trajectories, scalar DQN appeared to achieve an excellent failure rate of 0.67 HOF/min. However, when deployed on the unseen 1,000 test split, scalar DQN's failure rate surges by 67% to 1.12 ± 0.17 HOF/min—virtually matching the failure rate of the unlearned 3GPP baseline (1.15 ± 0.69 HOF/min).

This degradation exposes the fundamental vulnerability of scalar value learning in cellular mobility: because scalar DQN collapses returns into a single expected value $\mathbb{E}[R]$, the neural network overfits to the localized multipath fading profiles of the training set rather than learning invariant, generalizable channel dynamics.

In contrast, Risk-Neutral QR-DQN (RN QR-DQN) completely closes this generalization gap, reducing the out-of-domain failure rate to 0.77 ± 0.15 HOF/min. Requiring the network to predict $N = 25$ distinct quantiles across the full probability distribution forces the hidden representations to capture distribution-wide dispersion. This multi-target learning objective functions as a powerful implicit regularizer, rendering the agent robust to novel fading realizations.

B. Why Contextual Bandits Suffer Throughput Loss

Contextual bandits (CMAB) [5] provide strong safety protection, achieving 0.77 ± 0.45 HOF/min and the lowest ping-pong rate (1.3 PP/min). However, this protection comes at a heavy cost in spectral efficiency: CMAB achieves only 3.76 bps/Hz, suffering a severe 7.1% throughput penalty relative to deep RL (4.04 bps/Hz).

The origin of this penalty lies in the bandit's myopic objective ($\gamma = 0$). By treating each handover decision in isolation and assuming that current cell switches do not alter future state transitions, CMAB cannot anticipate future channel degradation along the trajectory. To avoid immediate execution failures, it conservatively remains on lower-throughput, robust links. Conversely, sequential deep RL agents ($\gamma = 0.9$) anticipate multi-step trajectory evolution, executing timely proactive handovers that maintain connectivity to high-order MCS sectors.

C. Failure of Truncated Risk (Study with $k \in \{1, 3, 7\}$) and CVaR Policy Paralysis

A crucial architectural question is whether risk-aware policies should restrict training capacity exclusively to the tail quantiles they consult at execution time. To address this, Table 2 includes the representative Truncated QR-DQN policy ($k = 7, \alpha = 0.25$), while we evaluate the broader parametric progression across $k = 1$ (RA-1q), $k = 3$ ($\alpha = 0.10$), and $k = 7$ ($\alpha = 0.25$) in the following breakdown:

Across all truncation levels, partial-tail training fails to generalize out-of-domain:

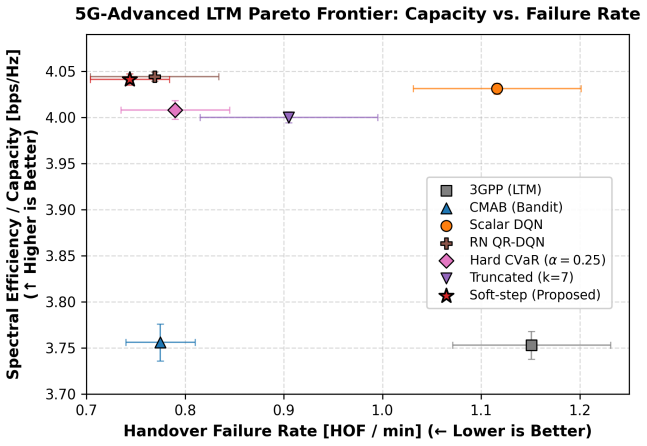


FIGURE 2. The 5G-Advanced LTM operating Pareto frontier: average spectral efficiency / capacity (bps/Hz) versus handover failure rate (HOF/min) across baseline and reinforcement learning policies on 1,000 unseen test trajectories (error bars indicate standard deviation across 5 independent random seeds). Soft-step QR-DQN establishes the optimal Pareto operating point, simultaneously maximizing capacity (4.04 bps/Hz) and cutting handover failures to the benchmark minimum (0.74 HOF/min) without the throughput loss of bandits or the mobility instability of scalar DQN.

- $k = 1$ (RA-1q): HOF rate degrades to 0.95 ± 0.16 HOF/min ($+27.5\%$ higher than Soft-step).
- $k = 7$ ($\alpha = 0.25$): HOF rate degrades to 0.90 ± 0.22 HOF/min ($+21.6\%$ higher than Soft-step).
- $k = 3$ ($\alpha = 0.10$): HOF rate is 0.84 ± 0.13 HOF/min, accompanied by a damaging drop in spectral efficiency down to 3.95 bps/Hz (a 2.4% loss in throughput).

This parametric study proves that the failure of truncated risk is structural rather than an artifact of single-quantile modeling. Restricting backpropagation exclusively to tail quantiles eliminates gradient feedback from median and upper return outcomes. Without full-distribution gradients, the trunk cannot learn robust features, leading to brittle tail-overfitting.

Furthermore, evaluating a full-distribution network under strict Hard CVaR with increased risk-aversion (corresponding to a smaller tail fraction $\alpha = 0.10$ vs. $\alpha = 0.25$) reveals a different pitfall: **policy paralysis** (as illustrated by the operating Pareto frontier in Fig. 2). At $\alpha = 0.10$, the agent becomes excessively risk-averse, reserving a peak 10.7% of network resources and triggering 39.2 preparations/min. Yet, out of fear of handover execution drops, it hesitates to execute necessary cell changes, allowing the serving link to deteriorate and paradoxically increasing failures to 0.83 HOF/min.

D. Superiority of Proposed Soft-Step QR-DQN

Soft-step QR-DQN ($\alpha = 0.25, r = 2.0$) resolves both the generalization gap and the policy paralysis dilemma:

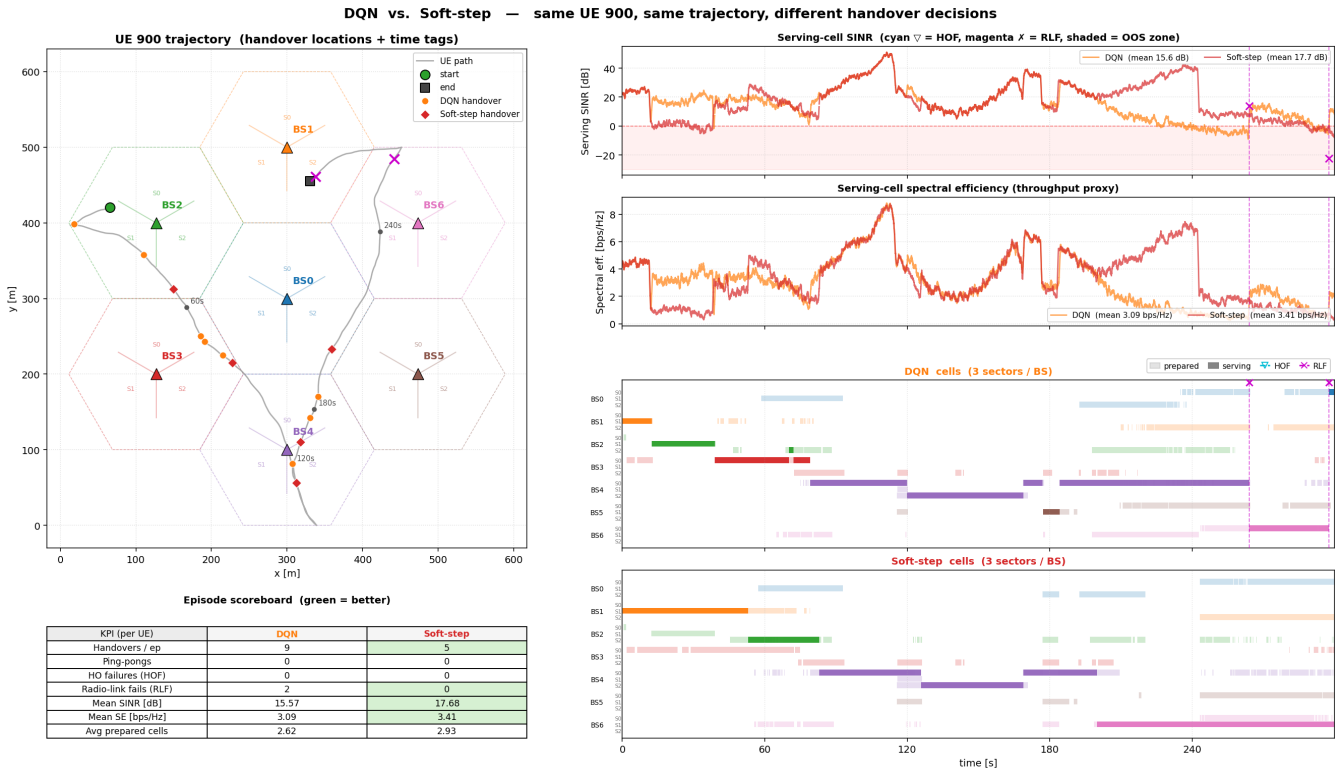
1) **Lowest Failure Rate in Benchmark:** Soft-step achieves $\mathbf{0.74 \pm 0.07}$ HOF/min, outperforming CMAB (0.77), RN QR-DQN (0.77), and all truncated variants.
2) **Halved Signaling Overhead:** Soft-step issues only 5.8 HO/min, cutting 3GPP control-plane signaling load (11.3 HO/min) by nearly 50%.
3) **Preserved Spectral Efficiency:** It achieves $\mathbf{4.04 \pm 0.01}$ bps/Hz, matching risk-neutral RL and completely avoiding the 7% capacity penalty of contextual bandits.
4) **Proximity to Operational Upper Bound:** Against a clairvoyant LTM-constrained execution oracle (which instantaneously selects the prepared candidate sector offering the highest SINR at each transmission interval with zero handover latency or execution failure, establishing an operational ceiling of 4.72 ± 0.27 bps/Hz, 94.4 Mbps), Soft-step captures 85.6% of the ideal realizable capacity (compared to 79.5% for 3GPP heuristics), closing 29.8% of the operational throughput gap.
5) **Unmatched Cross-Seed Robustness:** Soft-step cuts cross-seed failure variance by 53% compared to RN QR-DQN ($\sigma = 0.07$ vs. 0.15). By training on all 25 quantiles and applying a modest 2:1 smooth tail emphasis, Soft-step achieves robust risk-aversion without hyperparameter brittleness.

E. Temporal Dynamics: Mobile Trajectory Case Study (UE 900)

To provide microscopic insight into why Soft-step QR-DQN outperforms scalar DQN and eliminates handover failures, Fig. 3 presents a head-to-head temporal comparison across an identical 300 s mobile path from the unseen test set (UE 900). The user traverses the boundary between adjacent base stations, where severe shadow fading and multipath interference create rapid fluctuations in channel quality.

Under scalar DQN, the agent suffers from **handover procrastination**. As illustrated in the serving cell allocation raster (Fig. 3, bottom-right), DQN clings to BS 4 for over 150 s despite deteriorating radio conditions as the UE drives northeast toward BS 6. When localized fading drops push the serving link SINR below the out-of-sync threshold $Q_{\text{out}} = -3$ dB at $t \approx 265$ s and $t = 300$ s, DQN triggers two Radio Link Failures (RLFs), collapsing spectral efficiency to zero during connection recovery.

In contrast, Soft-step QR-DQN proactively executes timely, orderly sector transitions. By recognizing the accumulating tail probability of failure as the mobile departs BS 4, Soft-step hands over to BS 6 at $t \approx 200$ s, sustaining link continuity with higher SINR throughout the boundary traversal. As summarized in the trajectory scoreboard (Fig. 3, bottom-left), Soft-step achieves zero RLFs, improves average SINR by $+2.11$ dB (17.68 dB vs. 15.57 dB), and increases average spectral efficiency by 10.4% (3.41 bps/Hz vs. 3.09 bps/Hz) while reducing total handovers from 9 to 5.



KPI (per UE)	DQN	Soft-step
Handovers / ep	9	5
Ping-pongs	0	0
HO failures (HOF)	0	0
Radio-link fails (RLF)	2	0
Mean SINR [dB]	15.57	17.68
Mean SE [bps/Hz]	3.09	3.41
Avg prepared cells	2.62	2.93

FIGURE 3. **Head-to-head temporal trace on an identical out-of-domain mobile trajectory (UE 900, 300 s): (Top-left) Spatial trajectory and 7-site hexagonal base station deployment; (Top-right) Serving link SINR (dB) and spectral efficiency (bps/Hz); (Bottom-left) Trajectory KPI scoreboard summarizing cumulative handovers, radio link failures, mean serving SINR, and average spectral efficiency over the 300 s path; (Bottom-right) Serving cell allocation raster tracking the active serving sector index (0–20) over time for scalar DQN and proposed Soft-step QR-DQN. While scalar DQN procrastinates cell switching and triggers two connection drops (RLFs), Soft-step executes proactive, orderly sector transitions, achieving zero RLFs and higher effective capacity.**

VII. Conclusion and Outlook

In this paper, we investigated risk-aware distributional reinforcement learning for 5G-Advanced Lower-Layer Triggered Mobility (LTM). Under a rigorous out-of-domain evaluation on 1,000 mobile trajectories unseen during training, we uncovered a critical generalization gap in scalar DQN, whose failure rate nearly doubles relative to its in-sample training baseline due to overfitting local multipath fading profiles. We demonstrated that full-distribution Quantile Regression DQN (QR-DQN) provides an essential regularizing effect that restores out-of-domain generalization. Furthermore, our multi-quantile study revealed that truncating quantiles during training destroys generalization across all tail counts ($k \in \{1, 3, 7\}$), while strict Hard CVaR induces policy paralysis. To resolve these challenges, we proposed Soft-step QR-DQN ($\alpha = 0.25, r = 2.0$), which trains on the complete return distribution while smoothly upweighting the lower quartile. Across 9 standardized KPIs over 5 independent seeds, Soft-step achieves the lowest failure rate in the benchmark (0.74 HOF/min), cuts control-plane signaling in half, and preserves full spectral efficiency.

Limitations and Future Work. Our system model relies on standard 3GPP heuristic thresholds for candidate cell preparation and focuses optimization on the execution trigger. A promising future direction is extending the distributional framework to multi-agent or joint preparation-execution control, optimizing both candidate cell preparation and execution triggering simultaneously. Additionally, formulating the tail failure constraint via constrained MDPs (CMDP) or deploying the trained models onto open-source software-defined radio (SDR) testbeds represent vital avenues for experimental validation.

All simulation environments, training pipelines, model checkpoints, and mobile channel trajectory datasets are publicly accessible at [16] and [17].

Acknowledgment

This work covers activities of the project 6-SENSES grant PID2022-138648OB-I00, funded by MCIN/AEI/10.13039/501100011033, and by ERDF-EU. The authors also acknowledge the computational support and high-performance computing resources provided by the Universitat Politècnica de Catalunya – BarcelonaTech (UPC).

REFERENCES

[1] Ericsson, "5G Advanced Handover: L1/L2 Triggered Mobility," https://www.ericsson.com/en/blog/2024/8/5g-advanced-handover-triggered-mobility, 2024, ericsson Blog, August 2024.

[2] S. Alraih, I. Shayea, M. Behjati, S. J. Hashim, and A. Al-Fuqaha, "Revolutionizing handover management in 5G and beyond: A comprehensive survey of modern handover techniques, challenges and future directions," *IEEE Access*, vol. 10, pp. 90 547–90 579, 2022.

[3] V. Mnih, K. Kavukcuoglu, D. Silver, A. A. Rusu, J. Veness, M. G. Bellemare, A. Graves, M. Riedmiller, A. K. Fidjeland, G. Ostrovski, S. Petersen, C. Beattie, A. Sadik, I. Antonoglou, H. King, D. Kumaran, D. Wierstra, S. Legg, and D. Hassabis, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015.

[4] W. Dabney, M. Rowland, M. Bellemare, and R. Munos, "Distributional reinforcement learning with quantile regression," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, pp. 2892–2901, 2018.

[5] A. Y. Moreno-Locubiche, J. Vidal, O. Muñoz-Medina, and M. Cabrera-Bean, "Contextual bandits and reconfigurable intelligent surfaces for predictive LTM handover decisions," 2025. [Online]. Available: https://arxiv.org/abs/2512.08556

[6] M. G. Bellemare, W. Dabney, and R. Munos, "A distributional perspective on reinforcement learning," in *Proceedings of the 34th International Conference on Machine Learning*, ser. PMLR, vol. 70, 2017, pp. 449–458.

[7] W. Dabney, G. Ostrovski, D. Silver, and R. Munos, "Implicit quantile networks for distributional reinforcement learning," in *Proceedings of the 35th International Conference on Machine Learning*, ser. PMLR, vol. 80, 2018, pp. 1096–1105.

[8] D. Yang, L. Zhao, Z. Lin, T. Qin, J. Bian, and T.-Y. Liu, "Fully parameterized quantile function for distributional reinforcement learning," in *Advances in Neural Information Processing Systems*, vol. 32, 2019, pp. 1594–1604.

[9] C. Acerbi, "Spectral measures of risk: A coherent representation of subjective risk aversion," *Journal of Banking & Finance*, vol. 26, no. 7, pp. 1505–1518, 2002.

[10] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at risk," *Journal of Risk*, vol. 3, pp. 21–41, 2000.

[11] Y. Chow and M. Ghavamzadeh, "Algorithms for CVaR optimization in MDPs," *Advances in Neural Information Processing Systems*, vol. 27, 2014.

[12] A. Tamar, Y. Glassner, and S. Mannor, "Optimizing the CVaR via sampling," *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 29, no. 1, pp. 2993–2999, 2015.

[13] E. Eldeeb, H. Sifaou, O. Simeone, M. Shehab, and H. Alves, "Conservative and risk-aware offline multi-agent reinforcement learning," *IEEE Transactions on Cognitive Communications and Networking*, vol. 11, no. 3, pp. 1913–1926, 2025.

[14] E. Eldeeb and H. Alves, "Offline and distributional reinforcement learning for wireless communications," *IEEE Communications Magazine*, vol. 63, no. 8, pp. 71–76, 2025.

[15] J. Liu, D. Li, and Y. Xu, "Deep distributional reinforcement learning-based adaptive routing with guaranteed delay bounds," *IEEE/ACM Transactions on Networking*, vol. 32, no. 6, pp. 4692–4706, 2024.

[16] G. Grau, A. Y. Moreno-Locubiche, J. Vidal, and M. Cabrera-Bean, "Risk-Aware Learning for 5G-Advanced Handover Decisions (Source Code and Checkpoints)," https://github.com/gerardgrau/risk-aware-dist-rl-5g-handover, 2026.

[17] ——, "5G-Advanced LTM Mobile Channel Trajectories Dataset," https://doi.org/10.5281/zenodo.23124881, 2026.